

$$\frac{V}{2} \frac{\partial C}{\partial V} \cong \frac{M}{2} \frac{\partial C}{\partial M} \quad (7.6.32)$$

Applying Eq. (7.6.18) with Eqs. (7.6.25) through (7.6.30) and Eq. (7.6.32), for **high Reynolds number at the equilibrium reference state, the derivatives with respect to forward velocity** can be approximated as

$$\begin{aligned} F_{x_b,u} &\equiv \frac{\partial F_{x_b}}{\partial u} = \frac{\partial F_{x_b}}{\partial V} = \rho V_o S_w \left(C_X + \frac{M}{2} C_{X,M} \right) + \frac{\partial T}{\partial V} \cos(\alpha_{T0}) \\ &= -\frac{1}{2} \rho V_o^2 S_w \left(2 \frac{C_D}{V_o} + \frac{C_{D,M}}{a} \right) + \frac{\partial T}{\partial V} \cos(\alpha_{T0}) \\ F_{y_b,u} &\equiv \frac{\partial F_{y_b}}{\partial u} = \frac{\partial F_{y_b}}{\partial V} = \rho V_o S_w \left(C_Y + \frac{M}{2} C_{Y,M} \right) = 0 \\ F_{z_b,u} &\equiv \frac{\partial F_{z_b}}{\partial u} = \frac{\partial F_{z_b}}{\partial V} = \rho V_o S_w \left(C_Z + \frac{M}{2} C_{Z,M} \right) - \frac{\partial T}{\partial V} \sin(\alpha_{T0}) \\ &= -\frac{1}{2} \rho V_o^2 S_w \left(2 \frac{C_L}{V_o} + \frac{C_{L,M}}{a} \right) - \frac{\partial T}{\partial V} \sin(\alpha_{T0}) \quad (7.6.33) \\ M_{x_b,u} &\equiv \frac{\partial M_{x_b}}{\partial u} = \frac{\partial M_{x_b}}{\partial V} = \rho V_o S_w b_w \left(C_\ell + \frac{M}{2} C_{\ell,M} \right) = 0 \\ M_{y_b,u} &\equiv \frac{\partial M_{y_b}}{\partial u} = \frac{\partial M_{y_b}}{\partial V} = \rho V_o S_w \bar{c}_w \left(C_m + \frac{M}{2} C_{m,M} \right) + z_{T0} \frac{\partial T}{\partial V} \\ &= \frac{1}{2} \rho V_o^2 S_w \bar{c}_w \left(2 \frac{C_m}{V_o} + \frac{C_{m,M}}{a} \right) + z_{T0} \frac{\partial T}{\partial V} \\ M_{z_b,u} &\equiv \frac{\partial M_{z_b}}{\partial u} = \frac{\partial M_{z_b}}{\partial V} = \rho V_o S_w b_w \left(C_n + \frac{M}{2} C_{n,M} \right) = 0 \end{aligned}$$

where z_{T0} is the perpendicular offset between the thrust vector and the CG,

$$z_{T0} \equiv z_T \cos(\alpha_{T0}) + x_T \sin(\alpha_{T0})$$

The derivatives of the aerodynamic coefficients with respect to Mach number can be determined from wind tunnel data, which are generally gathered as a function of the Mach number. The derivatives are estimated from the data using finite difference. However, the required wind tunnel tests are time consuming and expensive. Thus, for preliminary design, it is convenient to have a first-order approximation for these derivatives that can be used to estimate their values prior to wind tunnel testing.

When the flow around an airfoil section remains subsonic, thin airfoil theory with the Prandtl-Glauert compressibility correction predicts a section lift coefficient given by

$$\tilde{C}_L = \frac{2\pi(\alpha - \alpha_{L0})}{\sqrt{1 - M^2}}$$